ಕರ್ನಾಟಕ ರಾಜ್ಯ ವಿಜ್ಞಾನ ಮತ್ತು ತಂತ್ರಜ್ಞಾನ ಮಂಡಳಿ

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50th Series of the STUDENT PROJECT PROGRAMME (SPP): 2026 - 2027

(Celebrating the Golden Jubilee of KSCST's flagship Student Project Programme)

Karnataka State Council for Science and Technology (KSCST) invites **project proposals** from **final-year students** of **B.E., B.Arch., M.Tech., M.Sc., M.Tech./M.Sc. (Agriculture), MBA, and MCA** programmes for sponsorship under the **50th Series of the Student Project Programme (SPP)**.

Online Submission Period: 21 September 2026 to 30 November 2026, 5.00 PM.
Project proposals shall be submitted only through the Google Form.

Suggested Broad Themes for Student Projects (SPP 50th Series)
High-Priority Focus Areas for Karnataka

The following broad themes are suggested for **B.E., B.Arch., M.Tech., M.Sc., M.Sc. (Agriculture), MCA and MBA** students. These themes align with Karnataka's development priorities, including agriculture, water, climate resilience, healthcare, digital transformation, manufacturing, renewable energy, circular economy and rural development, and are intended for projects of approximately **5 months duration**.

Broad Indicative Themes: The Project proposals should align with one of the following broad themes:

1) Sustainable Agriculture, Food Security, Rural Development and Inclusive Technologies
2) Water Resources, Environment and Climate Resilience
3) Renewable Energy, Bioenergy, Circular Economy, Waste Management and Green Technologies
4) Artificial Intelligence, Data Science and Digital Technologies
5) Electronics, Communication, IoT and Embedded Systems
6) Advanced Manufacturing, Robotics, Automation and Industry 4.0
7) Healthcare, Biotechnology and Biomedical Engineering
8) Sustainable Infrastructure, Smart Cities and Smart Mobility

9) Advanced Materials, Chemical Engineering and Sustainable Process Technologies
10) Natural Resources, Biodiversity, Forestry and Ecosystem Conservation
11) Governance, Public Policy, Education and Digital Society
12) Entrepreneurship, Innovation, Digital Business and Technology Management
13) Education, Skill Development and Digital Society
14) Disaster Risk Reduction, Resilience and Public Safety
15) Tourism, Heritage and Cultural Technologies
16) Aerospace, Space, Defence and Emerging Technologies
17) Multidisciplinary Technologies for Sustainable Livelihoods, MSME Development
18) Cross-cutting Technologies and Interdisciplinary Innovations
19) Biofuel / Bioenergy related projects

Broad Themes and Sub-Themes for the 50th Series Student Project Programme (SPP): 2026–2027

1. Sustainable Agriculture, Food Security, Rural Development and Inclusive Technologies

- Precision and Smart Agriculture
- Climate-Resilient Agriculture
- Digital Agriculture, Agri-Analytics and Decision Support Systems
- Protected Cultivation and Smart Irrigation
- Farm Mechanization and Agricultural Engineering
- Post-Harvest Technologies, Processing and Value Addition
- Agricultural Waste Utilization and Circular Agriculture
- Horticulture, Floriculture and Plantation Crops
- Livestock, Fisheries and Allied Agricultural Technologies
- Agri Supply Chain, Logistics and Market Intelligence
- Rural Livelihood Enhancement and Rural Entrepreneurship
- Smart Villages and Rural Infrastructure Development
- Rural Water Supply, Sanitation and Hygiene (WASH)
- Village Resource Mapping using GIS and Geospatial Technologies
- Affordable, Appropriate and Indigenous Technologies for Rural Communities
- Digital Inclusion and ICT-enabled Rural Services
- Women-Centric Technologies and Women Entrepreneurship
- Technologies for Tribal, Aspirational and Remote Regions
- Sustainable Natural Resource Management for Rural Development
- Climate-Smart and Sustainable Farming Systems

2. Water Resources, Environment and Climate Resilience

- Water Conservation, Management and Efficient Utilization
- Integrated River Basin, Watershed and Catchment Management

- Groundwater Assessment, Recharge and Aquifer Management
- Drinking Water Quality Monitoring, Purification and Treatment
- Wastewater Treatment, Recycling and Reuse
- Stormwater Management and Urban Flood Mitigation
- Flood, Drought and Disaster Risk Assessment and Management
- Climate Change Adaptation, Mitigation and Resilience
- Environmental Monitoring using IoT, GIS, Remote Sensing and Geospatial Technologies
- Air, Water and Soil Pollution Monitoring and Control
- Wetland, Lake, River and Ecosystem Conservation
- Coastal and Marine Environment Management
- Sustainable Water Supply, Sanitation and Hygiene (WASH)
- Environmental Impact Assessment and Natural Resource Management
- Carbon Footprint Assessment, Carbon Sequestration and Climate Action
- Nature-based Solutions for Environmental Sustainability
- Decision Support Systems for Water and Environmental Management
- Smart Technologies for Water, Environment and Climate Monitoring

3. Renewable Energy, Bioenergy, Circular Economy, Waste Management and Green Technologies

- Solar, Wind and Hybrid Renewable Energy Systems
- Biofuels, Bioenergy and Biomass Utilization
- Biomass Conversion Technologies (Biogas, Biochar, Briquetting, Gasification and Pyrolysis)
- Green Hydrogen and Alternative Clean Fuels
- Sustainable Aviation Fuel (SAF) and Advanced Biofuels
- Energy Storage Technologies and Battery Management Systems
- Smart Grids, Smart Microgrids and Distributed Energy Systems
- Electric Mobility, Electric Vehicles (EVs) and Charging Infrastructure
- Energy Efficiency, Energy Conservation and Energy Management
- Waste-to-Energy and Resource Recovery Technologies
- Solid, Plastic, Electronic (E-waste), Biomedical and Industrial Waste Management
- Circular Economy and Circular Manufacturing
- Sustainable Materials, Green Manufacturing and Eco-friendly Packaging
- Agricultural, Municipal and Industrial Waste Valorization
- Carbon Capture, Utilization and Storage (CCUS) and Carbon Reduction Technologies
- Carbon Footprint Assessment, Net-Zero and Decarbonization Strategies
- Environmental Sustainability and Green Technologies
- Sustainable Resource Recovery and Industrial Symbiosis
- AI, IoT and Digital Technologies for Energy and Waste Management
- Techno-economic Assessment and Life Cycle Analysis (LCA) of Renewable Energy and Bioenergy Systems

4. Artificial Intelligence, Data Science and Digital Technologies

- Artificial Intelligence (AI) and Machine Learning (ML)
- Data Science, Big Data Analytics and Business Intelligence
- Internet of Things (IoT) and Smart Connected Systems
- Robotics, Intelligent Automation and Autonomous Systems
- Computer Vision and Image Processing
- Natural Language Processing (NLP) and Generative AI
- Cyber Security, Information Security and Digital Forensics
- Cloud Computing, Edge Computing and Distributed Computing
- Digital Twins, Modelling and Simulation
- Blockchain, Web3 and Distributed Ledger Technologies
- Quantum Computing and Quantum Technologies
- Decision Support Systems and Intelligent Information Systems
- Human–Computer Interaction (HCI), Augmented Reality (AR), Virtual Reality (VR) and Extended Reality (XR)
- Software Engineering, Mobile Computing and Intelligent Application Development
- High-Performance Computing (HPC) and Scientific Computing
- AI, Data Analytics and Digital Technologies for Agriculture, Healthcare, Education, Industry and Governance

5. Electronics, Communication, IoT and Embedded Systems

- Internet of Things (IoT) and Smart Connected Systems
- Embedded Systems and Real-Time Applications
- VLSI Design, FPGA and System-on-Chip (SoC) Technologies
- Sensors, Sensor Networks and Instrumentation
- Signal, Image and Speech Processing
- Analog, Digital and Mixed-Signal Electronics
- Wireless, Optical and Satellite Communication Technologies
- 5G/6G, Next-Generation Networks and Internet Technologies
- Edge Computing, Fog Computing and Intelligent Edge Devices
- RF, Microwave and Antenna Engineering
- Consumer Electronics and Smart Electronic Systems
- Power Electronics, Embedded Control and Intelligent Energy Systems
- Electronic System Design, Testing and Reliability
- IoT, Embedded Systems and Communication Technologies for Agriculture, Healthcare, Industry and Smart Cities
- AI-enabled Embedded Systems and Intelligent Electronic Devices
- Cyber-Physical Systems and Industrial Internet of Things (IIoT)

6. Advanced Manufacturing, Robotics, Automation and Industry 4.0

- Robotics
- Industrial Automation
- Industry 4.0
- Smart Factories

- Additive Manufacturing
- CAD/CAM
- Process Optimization
- Quality Engineering
- Predictive Maintenance
- Advanced Manufacturing
- Additive Manufacturing (3D Printing)
- Industrial Automation
- Predictive Maintenance
- Smart Factories
- Quality Engineering
- Advanced Materials
- Industrial Safety Systems

7. Healthcare, Biotechnology and Biomedical Engineering

- Digital Healthcare, e-Health and Mobile Health (mHealth)
- Telemedicine, Telehealth and Remote Patient Monitoring
- Biomedical Engineering, Medical Devices and Healthcare Technologies
- Medical Imaging, Computer-Aided Diagnosis and AI in Healthcare
- Biotechnology, Molecular Biology and Applied Life Sciences
- Public Health, Epidemiology and Disease Surveillance Systems
- Healthcare Informatics, Electronic Health Records (EHR) and Health Information Systems
- Diagnostics, Point-of-Care Testing and Biosensors
- Drug Delivery Systems, Pharmaceutical Technologies and Biomaterials
- Assistive Technologies, Rehabilitation Engineering and Accessible Healthcare Solutions
- Medical Robotics, Wearable Health Devices and Smart Healthcare Systems
- Genomics, Bioinformatics and Precision Medicine
- Tissue Engineering, Regenerative Medicine and Biomaterials
- AI, IoT and Data Analytics for Healthcare and Biomedical Applications
- Healthcare Management, Hospital Information Systems and Digital Health Services
- Sustainable, Affordable and Inclusive Healthcare Technologies

8. Sustainable Infrastructure, Smart Cities and Smart Mobility

- Structural, Geotechnical and Earthquake Engineering
- Sustainable Infrastructure Planning, Design and Asset Management
- Smart Buildings, Green Buildings and Energy-Efficient Infrastructure
- Green Construction Materials and Sustainable Construction Technologies
- Construction Engineering, Project Management and Building Information Modelling (BIM)
- Transportation Engineering and Sustainable Mobility

- Smart Mobility, Intelligent Transport Systems (ITS) and Connected Mobility
- Electric Vehicles (EVs), Charging Infrastructure and Alternative Mobility Solutions
- Autonomous Vehicles, Drone Technologies and Unmanned Transportation Systems
- Railway, Metro and Public Transport Systems
- Urban and Regional Planning using GIS, Remote Sensing and Geospatial Technologies
- Infrastructure Health Monitoring, Structural Health Monitoring and Predictive Maintenance
- Disaster-Resilient, Climate-Resilient and Sustainable Infrastructure
- Smart Cities, Urban Services and Digital Urban Infrastructure
- Logistics, Supply Chain Management and Multimodal Transportation
- AI, IoT, Digital Twins and Data Analytics for Smart Infrastructure and Transportation

9. Advanced Materials, Chemical Engineering and Sustainable Process Technologies

- Advanced Functional, Smart and Intelligent Materials
- Nanotechnology and Nanomaterials
- Advanced Composites, Lightweight and High-Performance Materials
- Biomaterials, Biopolymers and Bio-based Materials
- Green Materials and Sustainable Material Technologies
- Energy Materials for Batteries, Fuel Cells and Hydrogen Technologies
- Advanced Ceramics, Glass and Electronic Materials
- Sustainable Chemical Processes and Green Chemistry
- Catalysts, Catalytic Processes and Process Intensification
- Membrane Technologies, Separation Processes and Water Treatment Materials
- Surface Engineering, Coatings and Corrosion Protection
- Circular Materials, Recycling and Resource Recovery Technologies
- Materials Characterization, Testing and Computational Materials Engineering
- AI, Modelling and Digital Technologies for Materials Design and Chemical Process Optimization
- Emerging Materials for Healthcare, Electronics, Aerospace and Clean Energy Applications
- Sustainable Manufacturing of Advanced Materials and Specialty Chemicals

10. Natural Resources, Biodiversity, Forestry and Ecosystem Conservation

- Natural Resource Assessment, Mapping and Sustainable Management
- Forest Resource Management, Afforestation and Forest Conservation
- Biodiversity Assessment, Conservation and Ecosystem Management
- Wildlife Conservation and Habitat Management
- Soil Health Assessment, Soil Conservation and Land Restoration
- Land Use Planning, Land Resource Management and Land Degradation Studies
- Wetland, Lake, River and Coastal Ecosystem Conservation
- Watershed Development and Integrated Natural Resource Management
- Ecosystem Restoration, Ecological Rehabilitation and Nature-based Solutions
- Mineral Resource Assessment, Sustainable Mining and Mine Rehabilitation
- Remote Sensing, GIS and Geospatial Technologies for Natural Resource Management

- Climate Change Impacts on Natural Resources and Ecosystem Resilience
- Carbon Sequestration, Ecosystem Services and Blue-Green Infrastructure
- Sustainable Utilization of Biological and Natural Resources
- Environmental Monitoring, Ecological Modelling and Decision Support Systems
- Community-Based Conservation, Eco-restoration and Sustainable Livelihoods

11. Governance, Public Policy, Education and Digital Society

- e-Governance and Digital Governance Platforms
- Smart Governance, Smart Panchayats and Smart Villages
- Citizen-Centric Digital Services and Public Service Delivery
- Public Policy, Policy Analysis and Good Governance
- Education Technologies, Digital Learning and Skill Development
- Digital Literacy, Digital Inclusion and Capacity Building
- Geographic Information Systems (GIS), Remote Sensing and Geospatial Technologies
- Decision Support Systems, Public Data Analytics and Business Intelligence
- Open Data, Digital Public Infrastructure and Data-Driven Governance
- Urban, Rural and Regional Planning using Geospatial Technologies
- Social Innovation, Community Development and Inclusive Governance
- Disaster Management, Early Warning Systems and Emergency Response Technologies
- AI, IoT and Digital Technologies for Governance and Public Administration
- Cyber Security, Privacy and Ethical Digital Governance
- Sustainable Development Planning and Monitoring using Geospatial and Digital Technologies
- Technology-enabled Governance for Agriculture, Healthcare, Education and Environmental Management

12. Entrepreneurship, Innovation, Digital Business and Technology Management

- Entrepreneurship, Startup Development and Business Incubation
- Innovation Management, Design Thinking and Product Development
- Technology Commercialization, Technology Transfer and Research Translation
- Digital Business, E-Commerce and Digital Business Models
- Business Analytics, Data-Driven Decision Making and Business Intelligence
- Financial Technologies (FinTech), Digital Payments and Financial Inclusion
- Social Entrepreneurship, Social Innovation and Inclusive Business Models
- Circular Economy, Sustainable Business Models and Green Entrepreneurship
- Intellectual Property Rights (IPR), Patents, Copyrights and Technology Licensing
- Technology Management, Research and Development (R&D) Management and Innovation Ecosystems
- Supply Chain Management, Operations Management and Logistics
- Marketing, Digital Marketing, Branding and Customer Relationship Management (CRM)
- Project Management, Strategic Management and Organizational Development
- MSME Development, Rural Entrepreneurship and Women Entrepreneurship
- AI, Data Analytics and Digital Technologies for Business Transformation

- Business Feasibility, Techno-economic Analysis and Sustainable Enterprise Development

13. Education, Skill Development and Digital Society

- Smart Learning Systems and Adaptive Learning Technologies
- Educational Technologies (EdTech) and Digital Learning Platforms
- Virtual Laboratories, Remote Laboratories and Simulation-based Learning
- Skill Development, Vocational Education and Workforce Readiness
- Digital Content Development, E-learning and Open Educational Resources (OER)
- Artificial Intelligence (AI), Learning Analytics and Intelligent Tutoring Systems
- Augmented Reality (AR), Virtual Reality (VR) and Extended Reality (XR) for Education
- Gamification, Immersive Learning and Interactive Educational Tools
- Digital Literacy, Information Literacy and Digital Inclusion
- Assistive Technologies and Inclusive Education for Persons with Disabilities
- STEM Education, Innovation and Experiential Learning
- Education Management Information Systems (EMIS) and Institutional Digital Transformation
- Online Assessment, Examination Systems and Academic Analytics
- Cyber Safety, Digital Ethics and Responsible Use of Technology in Education
- AI, IoT and Emerging Technologies for Education and Skill Development
- Lifelong Learning, Continuing Education and Capacity Building

14. Disaster Risk Reduction, Resilience and Public Safety

- Disaster Risk Assessment, Reduction and Resilience
- Early Warning Systems and Decision Support Systems
- Disaster Monitoring, Forecasting and Emergency Response Technologies
- Flood, Drought, Landslide, Cyclone and Earthquake Risk Assessment and Management
- Emergency Communication Systems and Crisis Management
- Fire Safety, Industrial Safety and Occupational Health
- Critical Infrastructure Protection and Resilient Infrastructure
- Community-Based Disaster Risk Management and Resilience Planning
- Remote Sensing, GIS and Geospatial Technologies for Disaster Management
- AI, IoT, Drones and Robotics for Disaster Preparedness and Response
- Climate Change Adaptation, Hazard Mitigation and Risk Reduction
- Search, Rescue and Humanitarian Technologies
- Public Health Emergency Preparedness and Pandemic Response
- Disaster Recovery, Rehabilitation and Sustainable Reconstruction
- Risk Communication, Public Awareness and Capacity Building
- Smart Technologies for Disaster Management and Public Safety

15. Tourism, Heritage and Cultural Technologies

- Heritage Conservation, Documentation and Digital Archiving
- Digital Museums, Virtual Museums and Cultural Informatics
- Sustainable Tourism, Eco-tourism and Responsible Tourism

- Heritage Tourism, Rural Tourism and Community-based Tourism
- GIS, Remote Sensing and Geospatial Technologies for Tourism Planning and Management
- Smart Tourism, Digital Tourism and Visitor Experience Technologies
- Augmented Reality (AR), Virtual Reality (VR) and Extended Reality (XR) for Heritage Interpretation
- Cultural Resource Management and Preservation of Traditional Knowledge
- Tourism Infrastructure Planning, Development and Management
- Tourism Information Systems, Mobile Applications and Digital Platforms
- AI, IoT and Data Analytics for Tourism and Heritage Management
- Accessibility, Inclusive Tourism and Universal Design
- Sustainable Destination Management and Environmental Conservation
- Promotion of Arts, Crafts, Handicrafts and Cultural Industries
- Tourism Entrepreneurship, Hospitality Management and Digital Marketing
- Decision Support Systems for Tourism Planning and Heritage Management

16. Aerospace, Space, Defence and Emerging Technologies

- Aerospace Engineering, Aerodynamics and Aircraft Systems
- Space Technologies, Satellite Systems and Space Exploration
- Unmanned Aerial Vehicles (UAVs), Drone Technologies and Autonomous Aerial Systems
- Remote Sensing, Earth Observation and Geospatial Applications
- Navigation, Positioning and Timing (PNT) Systems
- Aerospace Materials, Advanced Composites and Lightweight Structures
- Autonomous Systems, Intelligent Vehicles and Human–Machine Interaction
- Artificial Intelligence, Machine Learning and Robotics for Aerospace and Defence Applications
- Cyber Security, Secure Communication and Defence Information Systems
- Quantum Technologies, Quantum Computing and Quantum Communication
- Space-enabled Services for Agriculture, Disaster Management, Environment and Urban Planning
- Satellite Communication, Navigation and Earth Observation Applications
- Avionics, Flight Control Systems and Embedded Aerospace Systems
- Defence Technologies, Surveillance, Monitoring and Border Security Systems
- Simulation, Digital Twins and Modelling for Aerospace and Defence Systems
- Emerging Technologies for National Security, Resilience and Strategic Applications

17. Multidisciplinary Solutions for Sustainable Livelihoods and MSME Development

Projects addressing one or more of the Sustainable Development Goals (SDGs) through integrated and interdisciplinary approaches involving engineering, science, technology, management, architecture, agriculture, and social innovation.

Indicative Focus Areas:

- Smart Villages and Sustainable Rural Development

- Sustainable Campuses and Green Institutional Practices
- Climate-Smart Communities and Climate Resilience
- Inclusive, Affordable and Assistive Technologies
- Disaster Risk Reduction, Resilience and Community Preparedness
- Resource-Efficient, Low-Carbon and Circular Economy Solutions
- Artificial Intelligence (AI), IoT and Digital Technologies for Social Good
- Micro, Small and Medium Enterprises (MSME) Productivity Enhancement
- Rural Industrialization and Cluster Development
- Technologies for Traditional Industries, Artisans and Handicrafts
- Cooperative Development and Community-Based Enterprises
- Digital Marketing, E-Commerce and Market Access Platforms
- Sustainable Supply Chains and Value Chain Development
- Social Entrepreneurship and Inclusive Business Models
- Women and Youth Entrepreneurship, Skill Development and Employment Generation
- Local Resource-Based Technologies for Sustainable Livelihoods
- Technology-Enabled Financial Inclusion and Digital Empowerment
- Innovation for Aspirational Districts, Tribal Areas and Underserved Communities

18. Cross-cutting Technologies and Interdisciplinary Innovations

Projects integrating two or more disciplines to develop innovative, scalable and sustainable solutions for societal, industrial and environmental challenges.

Indicative Focus Areas:

- Artificial Intelligence (AI), Machine Learning (ML) and Data Analytics for Agriculture, Healthcare, Education and Governance
- Internet of Things (IoT), Sensor Networks and Smart Monitoring Systems
- GIS, Remote Sensing and Geospatial Technologies for Water, Environment, Climate and Urban Planning
- Robotics, Automation and Intelligent Systems for Healthcare, Manufacturing and Agriculture
- Digital Twins, Modelling and Decision Support Systems
- Smart Villages, Smart Campuses and Smart Community Solutions
- Green Buildings, Energy Management and Sustainable Infrastructure
- Bioenergy, Circular Economy and Resource-Efficient Technologies
- AI, IoT and Edge Computing for Industrial and Smart Manufacturing Applications
- Climate-Smart Technologies for Water, Agriculture and Disaster Resilience
- Assistive Technologies and Inclusive Solutions for Persons with Disabilities
- Drone Technologies, Autonomous Systems and Geospatial Applications
- Cyber-Physical Systems, Embedded Intelligence and Human–Machine Interaction
- Blockchain, Digital Platforms and Secure Data Management
- Sustainable Product Design, Frugal Innovation and Appropriate Technologies

- Technology Convergence for Sustainable Development Goals (SDGs) and Societal Impact

Discipline-wise Applicability

- **B.E. / B.Arch.:** Engineering Design, Prototypes, Embedded Systems, Automation, Infrastructure, Renewable Energy and Manufacturing.
- **M.Tech.:** Advanced Technologies, Modelling, Optimization, Simulation, AI, Industry 4.0 and Emerging Engineering Solutions.
- **M.Sc.:** Applied Sciences, Environmental Sciences, Biotechnology, Chemistry, Physics, Mathematics, Geospatial Sciences and Data Analytics.
- **MCA:** Software Systems, AI, Cyber Security, Mobile Applications, Cloud Computing, GIS, IoT and Digital Platforms.
- **MBA:** Technology Management, Innovation, Sustainability, Entrepreneurship, Supply Chains, Business Analytics, Finance, Marketing and Policy.

Note

These broad themes are consistent with Karnataka's Science, Technology and Innovation priorities, emphasizing agriculture, natural resources, health, energy efficiency, climate change, digital technologies, manufacturing and rural development, while also reflecting current thrust areas such as AI, renewable energy and advanced engineering.

The above themes are **indicative sub themes**. Institutions and students are encouraged to formulate innovative, interdisciplinary project proposals addressing local, regional or national challenges. Project topics should be well-defined and feasible for completion within an academic project duration of approximately **five months**.

Broad Theme	**Suitable 5-Month Project Examples**
Sustainable Agriculture	IoT-based soil moisture monitoring, AI-based crop disease detection, solar-powered irrigation controller
Water Resources	Low-cost water quality monitoring system, rainwater harvesting design for a campus, GIS mapping of village water bodies
Renewable Energy	Biomass briquette machine prototype, solar dryer, hybrid solar-biogas system, biochar production from agricultural waste
Circular Economy	Plastic waste-based paver blocks, composting system for institutions, food waste biogas unit
Healthcare	Smart medicine reminder, AI-based diabetic retinopathy screening, portable health monitoring device
AI & Data Science	Attendance analytics, traffic prediction, crop yield prediction, chatbot for citizen services
Smart	Automated sorting system, predictive maintenance model, IoT energy

Manufacturing	monitoring in industries
Smart Mobility	EV charging station management, smart parking system, accident detection system
Smart Cities	Smart street lighting, air quality monitoring network, smart waste bin
Rural Development	Digital marketplace for farmers, village GIS mapping, low-cost farm tools
Disaster Management	Flood warning prototype, landslide susceptibility mapping, emergency communication app
Tourism	Heritage mapping using GIS, AR tourism guide, tourism recommendation system
Bioenergy	Biodiesel from waste cooking oil, biogas yield optimization, biomass characterization, biochar production
Entrepreneurship	Business model for biomass supply chain, EV charging startup feasibility, carbon credit business model

KSCST Encourages interdisciplinary projects instead of forcing students into rigid departments, projects like:

- AI + Agriculture
- GIS + Water
- IoT + Healthcare
- Bioenergy + Circular Economy
- AI + Manufacturing

Can be proposed. Projects addressing societal challenges, sustainability, rural development, climate resilience, digital transformation, and indigenous technologies through innovative, multidisciplinary approaches are particularly encouraged.

19. Broad Themes in Biofuel / Bioenergy area

1. Biomass Resource Assessment and Characterization

- Assessment of locally available biomass resources for bioenergy production.
- Characterization of agricultural residues and forest biomass.
- Biomass availability mapping using GIS and remote sensing.
- Seasonal biomass estimation and supply chain analysis.

2. Biofuel Production Technologies

- Biodiesel production from non-edible oils and waste cooking oil.
- Bioethanol production from agricultural residues and food waste.
- Biogas production from organic wastes.
- Biohydrogen production using biological processes.
- Sustainable Aviation Fuel (SAF) feedstock assessment.

3. Biomass Conversion Technologies

- Briquetting and pelletization of biomass residues.
- Torrefaction and carbonization of biomass.
- Pyrolysis and gasification studies.
- Production of biochar from agricultural wastes.
- Comparative analysis of biomass conversion technologies.

4. Waste-to-Energy Technologies

- Energy recovery from municipal organic waste.
- Biogas generation from institutional and community waste.
- Anaerobic digestion optimization.
- Food waste valorization for bioenergy.
- Integrated waste management for energy generation.

5. Algal Biofuels

- Cultivation of microalgae for biofuel applications.
- Lipid extraction techniques.
- Wastewater-based algal cultivation.
- Algal biomass characterization.
- Techno-economic feasibility of algal biofuel production.

6. Bioenergy for Rural Development

- Village-scale bioenergy systems.
- Biomass-based decentralized power generation.
- Bioenergy for rural livelihood enhancement.
- Clean cooking technologies using biomass.
- Community biogas models.

7. Biomass Supply Chain and Logistics

- Biomass collection and transportation optimization.

- GIS-based biomass supply chain planning.
- Digital platforms for biomass aggregation.
- Cost analysis of biomass logistics.

8. Bioenergy and Circular Economy

- Agricultural waste valorization.
- Integrated bio-refinery concepts.
- Circular bioeconomy models.
- Resource recovery from agro-industrial waste.

9. Bioenergy and Climate Change

- Carbon footprint analysis of biofuels.
- Life Cycle Assessment (LCA) of bioenergy systems.
- Carbon sequestration potential of biochar.
- GHG mitigation through biomass utilization.

10. Bioenergy Policy and Economics

- Economic feasibility of biomass energy projects.
- Policy analysis for biofuel adoption.
- Techno-economic assessment of bioenergy systems.
- Biofuel market potential and business models.

11. Bioenergy Equipment and Systems

Suitable for Mechanical, Electrical and Electronics students:

- Design of biomass dryers.
- Low-cost biomass briquetting machines.
- Smart biomass cookstoves.
- Automated biomass feeding systems.
- IoT-enabled biogas plant monitoring.
- Biomass combustion efficiency improvement.

12. Artificial Intelligence and Digital Technologies for Bioenergy

- AI-based biomass yield prediction.
- Machine learning models for biogas production.
- IoT-based monitoring of biodigesters.
- Digital twin for bioenergy systems.
- Remote sensing for biomass estimation.

13. Sustainable Agriculture and Bioenergy

- Energy crops for marginal lands.
- Crop residue management for bioenergy.
- Integrated farming and bioenergy systems.
- Bioenergy from livestock waste.
- Biomass production optimization.

14. Bioenergy from Industrial Residues

- Distillery waste utilization.
- Sugar industry residues.
- Paper and pulp waste.
- Textile organic waste.
- Food processing residues.

15. Emerging Bioenergy Technologies

- Bio electrochemical systems.
- Microbial fuel cells.
- Synthetic biology for biofuels.
- Green hydrogen from biomass.
- Hybrid renewable energy systems integrating biomass.

Suggested Themes by Discipline

B.E. (Engineering)

- Biomass processing equipment
- Smart bioenergy systems
- IoT and automation for biogas plants
- Waste-to-energy technologies
- Biomass gasifiers
- Briquetting and pelletization
- Energy-efficient biomass combustion
- AI applications in biomass resource management

M.Tech.

- Advanced biomass conversion technologies
- Process optimization using modelling and simulation
- Bio-refinery design
- Techno-economic analysis
- Life Cycle Assessment (LCA)
- Computational modelling of bioenergy systems
- Hybrid renewable energy systems

M.Sc.

- Microbial biofuel production
- Algal biofuels
- Feedstock characterization
- Biochar properties
- Biogas microbiology
- Biomass quality assessment
- Enzyme-assisted biomass conversion

- Environmental impact assessment

High-Priority Focus Areas for Karnataka

Considering Karnataka's agricultural and industrial profile, the following areas are relevant:

- Sugarcane trash and bagasse utilization
- Paddy straw and rice husk bioenergy
- Coconut shell and coir waste utilization
- Arecanut and coffee residues
- Maize stalk and cobs
- Sericulture waste
- Horticultural residues
- Temple flower waste
- Municipal biodegradable waste
- Dairy and poultry waste
- Decentralized village bioenergy systems
- Bio-CNG and compressed biogas (CBG)
- Biochar for soil health and carbon sequestration
- Sustainable Aviation Fuel (SAF) feedstocks
- AI and IoT applications in biomass supply chains and bioenergy plants